HOL1: Baseline or Bust - Baseline Report

Source run: mock_baseline.csv | model: MOCK | variant: baseline | generated 2026-10-04

SAMPLE LAYOUT - SYNTHETIC MOCK DATA, NOT A REAL BENCHMARK. Do not submit.

Executive Summary

Latency is tightly distributed (total p50 1554 ms, p95 2339 ms, ratio 1.51x; TTFT p50 311 ms). Throughput averaged 80.5 tokens/sec and cost averaged $0.000084 per task ($0.00126 for 15 prompts). 0 latency and 0 cost outlier(s) were flagged; the slowest requests (listed above) are the tail to watch, not a clear bottleneck. All requests succeeded, so the baseline is a usable reference for optimization experiments.

Summary Table

Metric	Value
p50 ttft_ms	310.8
p95 ttft_ms	467.9
p50 total_ms	1554.2
p95 total_ms	2339.4
p95/p50 ttft	1.51x
p95/p50 total	1.51x
avg tokens_per_sec	80.5
avg cost_per_task (USD)	0.000084
total cost, all tasks (USD)	0.001264
avg input / output tokens	25 / 134
latency outliers (Tukey fence)	0 (> 3323 ms)
cost outliers (Tukey fence)	0
successful requests	15/15

Outliers: slowest requests (by total_ms)

prompt_id	prompt	total_ms	out_tok	cost
4	Mock prompt 4	2343	211	0.000131
8	Mock prompt 8	2338	210	0.000130
11	Mock prompt 11	2101	196	0.000122

Outliers: highest cost

prompt_id	prompt	cost	out_tok
4	Mock prompt 4	0.000131	211
8	Mock prompt 8	0.000130	210

prompt_id	prompt	cost	out_tok
11	Mock prompt 11	0.000122	196

Reflection: Two Hypotheses

Hypothesis 1 - output length drives latency and cost. Output tokens correlate with total_ms at r = 1.00; responses longer than the median (125 tokens) average 2071 ms versus 1198 ms for shorter ones, and the slowest request (prompt 4, 2343 ms) produced 211 tokens. Output tokens are priced 4x input, so a tighter max_tokens or a 'be concise' instruction should cut both total_ms and cost_per_task (avg 0.000084 USD) with little loss for simple support questions. I would test this by re-running the same 15 prompts with max_tokens halved and comparing p95 total_ms and cost.

Hypothesis 2 - first-token wait is a separate cost worth attacking on its own. p50 ttft_ms is 311 ms with p95/p50 = 1.51x, and TTFT is a median 20% of total_ms. The ratio is below 3x, so jitter is modest; gains must come from shortening the prompt/prefill, caching the shared system prompt, or routing simple FAQ-style prompts to a smaller model. Success means a lower p50 ttft_ms without raising p95.

Per-prompt Baseline (all test cases)

prompt_id	ttft_ms	total_ms	input_tokens	output_tokens	tokens_per_sec	cost_per_task
1	310.833	1554.17	24	125	80.429	7.9e-05
2	322.306	1611.53	21	132	81.9097	8.32e-05
3	395.455	1977.27	29	175	88.5057	0.000109
4	468.547	2342.74	27	211	90.0656	0.0001306
5	151.017	755.084	29	46	60.9204	3.16e-05
6	194.393	971.967	20	65	66.8747	4.3e-05
7	410.686	2053.43	27	188	91.5541	0.0001168
8	467.582	2337.91	22	210	89.8239	0.00013
9	231.836	1159.18	25	84	72.4652	5.44e-05
10	248.121	1240.6	29	96	77.3817	6.16e-05
11	420.289	2101.45	22	196	93.2691	0.0001216
12	284.685	1423.42	27	116	81.4936	7.36e-05
13	240.33	1201.65	21	89	74.0647	5.74e-05
14	413.99	2069.95	23	188	90.8234	0.0001168
15	255.505	1277.53	29	86	67.3177	5.56e-05